

Sponsors

R Ignite 2026 Workshop

September
11 – 12



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R IGNITE 2026

Introduction to R and RStudio

Session 1 • September 11, 2026 • 9:30–10:30 a.m.

USIU-Africa School of Pharmacy and Health Sciences
Nairobi, Kenya



Objectives of Session 1

Note: No prior experience with R/RStudio is necessary during this session

By the end, you can

- Explain R vs RStudio
- Navigate the main panes
- Run simple commands
- Create and inspect objects

And you will have

- A small health dataset
- Simple descriptive statistics
- One visualization
- A saved, reproducible script



Why R for health research?

Research questions

- Who is at greatest risk for cervical cancer?
- How do outcomes differ by group?
- What patterns are present in surveillance data?
- How uncertain are our estimates?

R helps us

- Manage and clean data
- Analyze and model
- Visualize and communicate
- Reproduce the full workflow

R and RStudio are not the same thing

R

The engine
Programming language +
statistical computing



RStudio

The dashboard
Editor + console + plots + files +
packages

Install R + RStudio Desktop

Do this before the hands-on lab • R is installed first, then RStudio

1 Go to Posit Downloads

posit.co/downloads

2 Install R first

Choose the link for your operating system and install the current R release.

3 Download RStudio Desktop

Choose the free Open Source edition for Windows, macOS, or Linux.

4 Run the installer

Accept the standard/default options unless your institution specifies otherwise.

5 Launch RStudio

RStudio should detect your installed R automatically. You are ready when the Console shows a > prompt.

Quick installation check

In the RStudio Console, type:

```
R.version.string
```

If a version is returned, R and RStudio are communicating correctly.

Current RStudio Desktop release in Posit documentation:
2026.08.2

Tip: If installation is blocked on a university-managed laptop, ask IT for administrator support.

RStudio after installation: the four (**paneli**) panes

A simplified view of the default RStudio workspace

RStudio	
SOURCE	ENVIRONMENT
Write & save scripts File: analysis.R	See objects that you created in memory age 35 bmi 28.4
CONSOLE	OUTPUT
Run command and see output > 2+3 [1] 5 > mean(c(2,4,6)) [1] 4	Files Plots Packages Help Plots and other outputs appear here.

What each pane does

1 Source

Create and save R scripts, R Markdown, or Quarto documents.

2 Console

Run commands interactively and see immediate output.

3 Environment

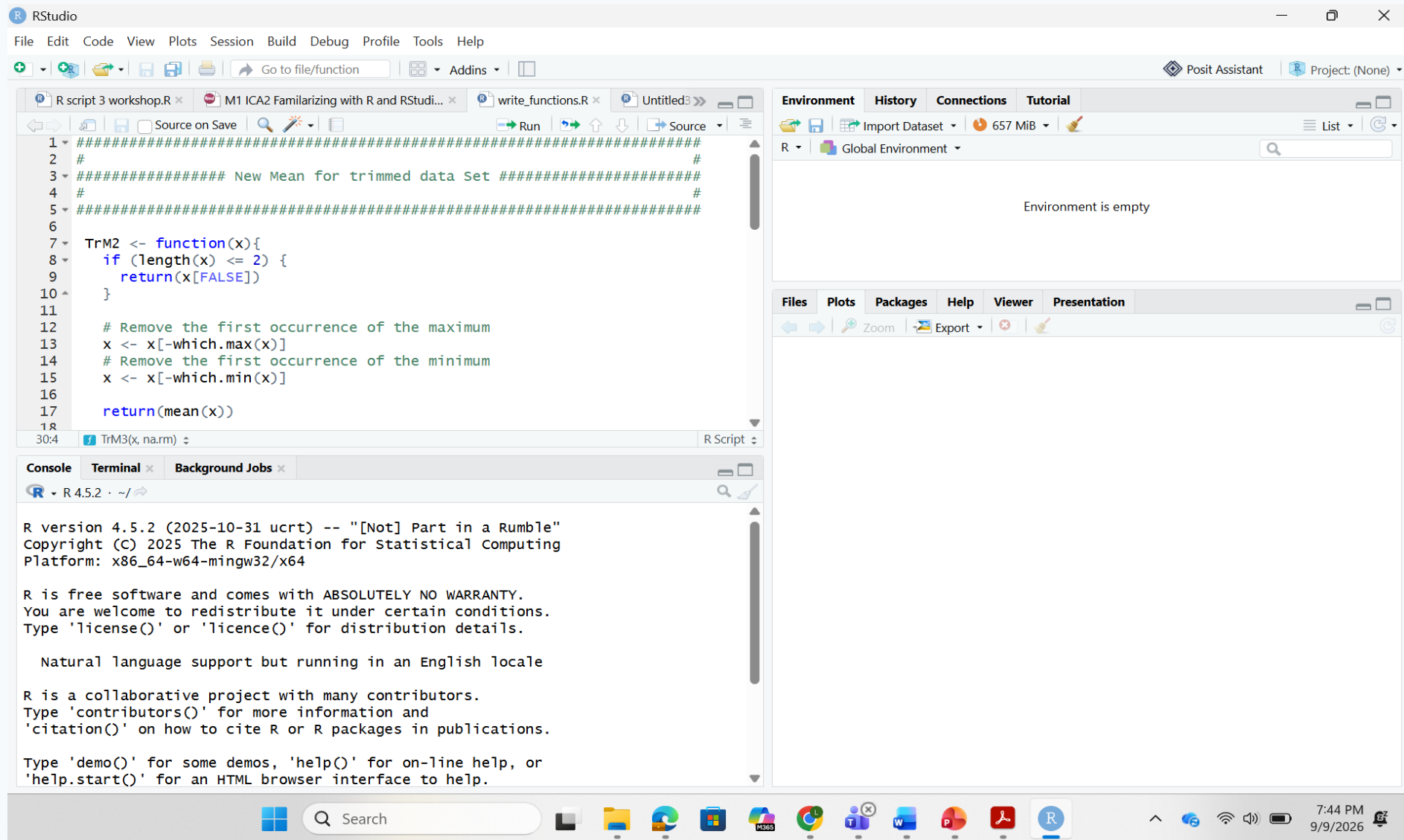
See objects created during the current R session.

4 Output

Access Files, Plots, Packages, Help, Viewer, and more.

Pane locations can be customized later in Tools → Global Options → Pane Layout.

RStudio: the actual four panes



Your first R commands

Type along in the Console

```
2 + 3  
20 / 4  
sqrt(25)
```

R evaluates expressions and returns results.

Try changing one number.

Mistakes are normal—error messages are information.

Objects: give information a name

```
age <- 35
```

```
age
```

```
bmi <- 28.4
```

```
bmi
```

Read <- as “assign to”.

- Object names should be meaningful
- R is case-sensitive: age ≠ Age
- Run a script line with Ctrl/Cmd + Enter

Vectors: several values in one object

```
blood_pressure <- c(120, 135, 140, 128)
```

```
mean(blood_pressure)
```

```
sd(blood_pressure)
```

Health data are full of vectors:

- ages
- BMI values
- blood pressures
- lab measurements

Common data types

Numeric

45, 28.4

Character

"Female"

Logical

TRUE / FALSE

Factor / category

Treatment group

The variable type affects what analyses and graphs make sense.

A health data frame

Rows = participants • Columns = variables

ID	Age	BMI	Diabetes
1	45	28	No
2	50	32	Yes
3	35	24	No
4	60	35	Yes
5	42	27	No

Think like a researcher:

- What is one row?
- What does each column mean?
- Which variables are numeric?
- Which are categorical?

Inspect before you analyze

```
View(patient_data)  
dim(patient_data)  
names(patient_data)  
str(patient_data)  
summary(patient_data)
```

A good first habit:

Look at the structure before choosing a statistical method.

Packages extend R

Install once

```
install.packages("ggplot2")
```

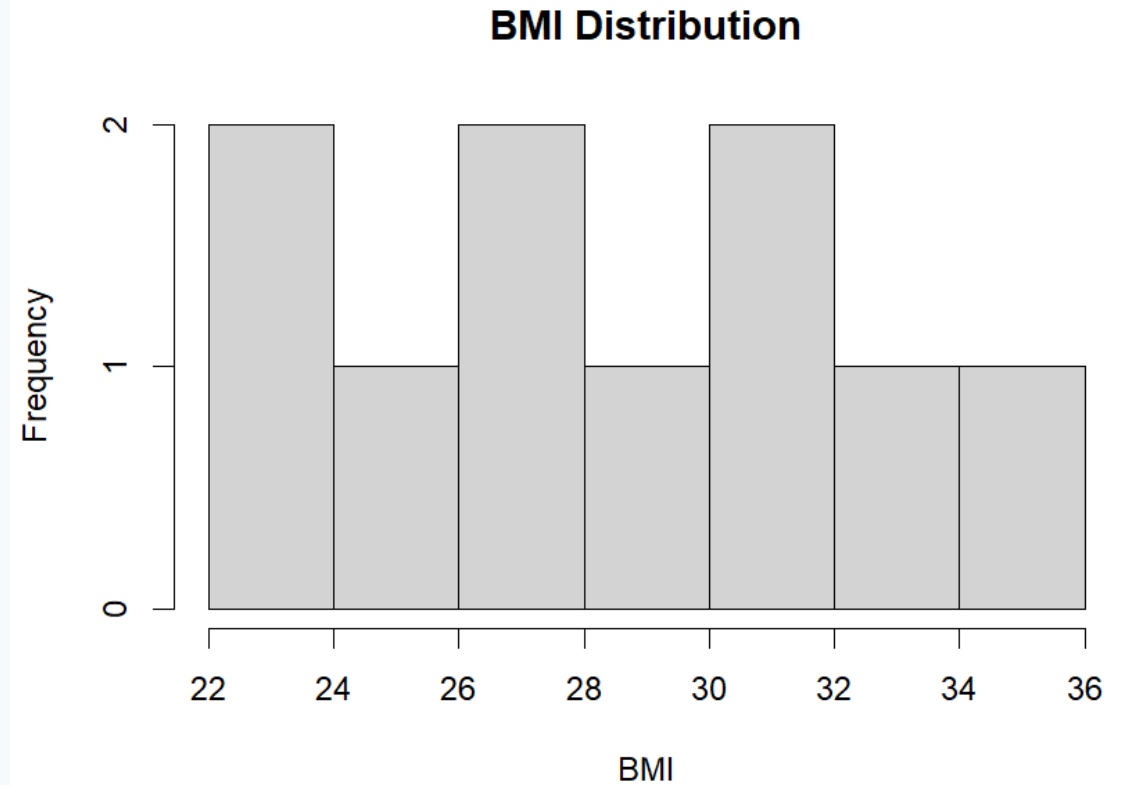
Load each session

```
library(ggplot2)
```

Examples you will encounter in this workshop and beyond: tidyverse • ggplot2 • epiR •
Bayesian packages

Your first visualization

```
hist(patient_data$BMI,  
      main = 'BMI Distribution',  
      xlab = 'BMI')
```



What does the distribution suggest?

Reproducibility: save the recipe, not just the result



Project/
data/
scripts/
results/
reports/

Mini challenge: 3 minutes

Create a vector of five BMI values and answer:

- What is the mean BMI?
- What is the standard deviation?
- What is the median?
- Change one value and rerun the commands. What changed?

When R gives an error

Errors are part of programming—not evidence that you “cannot code.”

```
mean(patient_data$bmi)
```

Possible message: object/column not found

Debugging routine

- Read the message
- Check spelling and capitalization
- Check parentheses and quotes
- Check that the object was created

Quick confidence check

I can now...

- Open and navigate RStudio
- Run a command
- Create an object
- Inspect a data frame
- Find a plot

Rate yourself

1 2 3 4 5

very low

very high

Next: Session 2 — Interactive R Primer and Exercises

You now know the workspace and basic grammar.

After the health break, we move from “watch and try” to extended hands-on practice with health data.

Bring your laptop + your questions